In [1]:
```
import csv
import decimal

import numpy as np
import pandas as pd
```

In [2]:
```
a = np.array([1, 2, 3])
print(a)
print(a.dtype)
```

```
[1 2 3]
int64
```

In [3]:
```
a = np.array([1, 2, 3], dtype = np.int16)
print(a)
print(a.dtype)
```

```
[1 2 3]
int16
```

In [4]:
```
np.random.seed(0)
print(np.random.random(5))
```

```
[0.5488135  0.71518937 0.60276338 0.54488318 0.4236548 ]
```

In [5]:
```
print(np.array(range(1, 7)).reshape(3, 2))
```

```
[[1 2]
 [3 4]
 [5 6]]
```

In [6]:
```
a = np.array([1, 2, 3, 4, 5, 6])
print(a)
```

```
[1 2 3 4 5 6]
```

In [7]:
```
b = a.reshape(2, 3)
print(b)
```

```
[[1 2 3]
 [4 5 6]]
```

In [8]:
```
b[0][0] = 10
print(b)
print(a)
```

```
[[10  2  3]
 [ 4  5  6]]
[10  2  3  4  5  6]
```

In [9]:
```
c = a.reshape(2, 3).copy()
print(c)
c[0][0] = 100
```

```
print(c)
print(a)
```

```
[[10  2  3]
 [ 4  5  6]]
[[100   2   3]
 [  4   5   6]]
[10  2  3  4  5  6]
```

In [10]:

```
a = np.arange(10).reshape(2, 5)
b = np.arange(11, 21).reshape(2, 5)
print(a)
print(b)
```

```
[[0 1 2 3 4]
 [5 6 7 8 9]]
[[11 12 13 14 15]
 [16 17 18 19 20]]
```

In [11]:

```
c = np.vstack((a, b))
print(c)
```

```
[[ 0  1  2  3  4]
 [ 5  6  7  8  9]
 [11 12 13 14 15]
 [16 17 18 19 20]]
```

In [12]:

```
c = np.vstack([a, b])
print(c)
```

```
[[ 0  1  2  3  4]
 [ 5  6  7  8  9]
 [11 12 13 14 15]
 [16 17 18 19 20]]
```

In [13]:

```
a = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print(a)
print(a.shape)
print(a.size)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
(3, 3)
9
```

In [14]:

```
b = np.array([10, 20, 30])
print(b.size)
```

```
3
```

In [15]:

```
a[::, 1] = b
print(a)
```

```
[[ 1 10  3]
 [ 4 20  6]
 [ 7 30  9]]
```

In [16]:
```
a = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print(a)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

In [17]:
```
a[::, ::] = b
print(a)
```

```
[[10 20 30]
 [10 20 30]
 [10 20 30]]
```

In [18]:
```
a = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print(a)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

In [19]:
```
a = np.array([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12]])
print(a)
```

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
```

In [20]:
```
print(a[1, 3])
print(a[np.array([1]), np.array([2])])
print(a[np.array(1), np.array(2)])
print(a[np.array([0, 2]), np.array([2, 3])])
print(a[np.array([1]), [2]])
try:
        print(a[np.array(1, 2)])
except TypeError as ex:
        print(type(ex).         + ": " + ex.args[0])
print(a[np.array([1, 2])])
print(a[(0, 1), (2, 3)])  # [(0, 2), (1, 3)]
print(a[np.array([0, 1]), np.array([2, 3])])  # [(0, 2), (1, 3)]
print(a[[0, 1], [2, 3]])
```

```
8
[7]
7
[ 3 12]
[7]
TypeError: Cannot interpret '2' as a data type
[[ 5  6  7  8]
 [ 9 10 11 12]]
[3 8]
[3 8]
[3 8]
```

In [21]:
```
a = np.arange(10, 100, 10)
print(a)
```

```
[10 20 30 40 50 60 70 80 90]
```

In [22]:
```
b = np.array([a[0], a[2], a[3]])
print(b)
```

```
[10 30 40]
```

In [23]:
```
c = a[np.array([0, 2, 3])]
print(c)
```

```
[10 30 40]
```

In [24]:
```
try:
        print(a[0, 2, 3])
except IndexError as ex:
        print(type(ex).         + ": " + ex.args[0])
```

```
IndexError: too many indices for array: array is 1-dimensional, but 3 were indexed
```

In [25]:
```
print(a[[0, 2, 3]])
```

```
[10 30 40]
```

In [26]:
```
try:
        print(a[(0, 2, 3)])
except IndexError as ex:
        print(type(ex).         + ": " + ex.args[0])
```

```
IndexError: too many indices for array: array is 1-dimensional, but 3 were indexed
```

In [27]:
```
try:
        a = np.array([[1], [2, 3], [4, 5, 6]])
        print(a)
except ValueError as ex:
        print(type(ex).         + ": " + ex.args[0])
```

```
ValueError: setting an array element with a sequence. The requested array has an inhomogeneous shape
after 1 dimensions. The detected shape was (3,) + inhomogeneous part.
```

In [28]:
```
print(np.zeros((2, 3), dtype = np.int8))
print(np.zeros([2, 3], dtype = np.int8))
```

```
[[0 0 0]
 [0 0 0]]
[[0 0 0]
 [0 0 0]]
```

In [29]:
```
print(np.eye(2, dtype = np.int8))
```

```
[[1 0]
 [0 1]]
```

In [30]:
```
print(np.arange(2, 8))
```

```
[2 3 4 5 6 7]
```

In [31]:
```
print(np.linspace(2, 5, 4, dtype = np.uint8))
```

```
[2 3 4 5]
```

In [32]:
```
print(np.random.random())
print(np.random.random(2))
print(np.random.random((2, 3)))
try:
        print(np.random.random(4, (2, 3)))
except TypeError as ex:
        print(type(ex).          + ": " + ex.args[0])
```

```
0.6458941130666561
[0.43758721 0.891773  ]
[[0.96366276 0.38344152 0.79172504]
 [0.52889492 0.56804456 0.92559664]]
TypeError: random() takes at most 1 positional argument (2 given)
```

In [33]:
```
try:
        print(np.random.randint())
except TypeError as ex:
        print(type(ex).          + ": " + ex.args[0])
print(np.random.randint(20))
print(np.random.randint((2, 3)))  # Two random integers with the first < 2 and second < 3 (different upper bounds).
try:
        print(np.random.randint(4, (2, 3)))
except ValueError as ex:
        print(type(ex).          + ": " + ex.args[0])
print(np.random.randint(10, size = (2, 3)))  # 2x3 matrix with high = 10, half-open, remember ?
```

```
TypeError: randint() takes at least 1 positional argument (0 given)
13
[0 1]
ValueError: low >= high
[[4 3 0]
 [3 5 0]]
```

In [34]:
```
a = np.array([np.arange(1, 6), np.arange(6, 11), np.arange(11, 16),
np.arange(16, 21), np.arange(21, 26)])
print(a)
```

```
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]
 [11 12 13 14 15]
 [16 17 18 19 20]
 [21 22 23 24 25]]
```

In [35]:

```
b = a[::2, 1::2]
print(b)
c = [[20, 40], [120, 140], [220, 240]]
a[::2, 1::2] = c
print(a)
print(a[3, 1::2])
```

```
[[ 2  4]
 [12 14]
 [22 24]]
[[  1  20   3  40   5]
 [  6   7   8   9  10]
 [ 11 120  13 140  15]
 [ 16  17  18  19  20]
 [ 21 220  23 240  25]]
[17 19]
```

In [36]:

```
a = np.array([1, 2, 3, 4, 5, 6])
b = np.array([10, 20])
a[:2] = b
print(a)
print(b)
a[0] = 100
print(a)
print(b)
b[0] = 1000
print(a)
print(b)
```

```
[10 20  3  4  5  6]
[10 20]
[100  20   3   4   5   6]
[10 20]
[100  20   3   4   5   6]
[1000   20]
```

In [37]:

```
print(np.array([[1], [2]]))
```

```
[[1]
 [2]]
```

In [38]:

```
a = np.array([np.arange(1, 6), np.arange(6, 11), np.arange(11, 16),
np.arange(16, 21), np.arange(21, 26)])
print(a)
```

```
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]
 [11 12 13 14 15]
 [16 17 18 19 20]
 [21 22 23 24 25]]
```

In [39]:
```
print(a[np.array([[1], [2]])])
print(a[np.array([1, 2])])
print(a[1, 2])
```

```
[[[ 6  7  8  9 10]]

 [[11 12 13 14 15]]]
[[ 6  7  8  9 10]
 [11 12 13 14 15]]
8
```

In [40]:
```
a = np.arange(10, 100, 10)
print(a)
b = a[np.array([0, 1, 1, 5])]
print(b)
print(a[np.array([[0, 1], [1, 5]])])
```

```
[10 20 30 40 50 60 70 80 90]
[10 20 20 60]
[[10 20]
 [20 60]]
```

In [41]:
```
a = np.array([[1, 2], [3, 4]])
print(a[a != 3])
```

```
[1 2 4]
```

In [42]:
```
a = [[1, 2], [3, 4]]
print(a[a != 3])
```

```
[3, 4]
```

In [43]:
```
a = [[1, 2], [3, 4]]
b = a[a != 3]
print(b)
```

```
[3, 4]
```

In [44]:
```
a = [[1, 2], [3, 4]]
mask = a != 3
print(mask)
```

```
True
```

In [45]:
```
a = np.array([[1, 2], [3, 4]])
mask = a != 3
print(mask)
```

```
[[ True  True]
 [False  True]]
```

In [46]:
```
a = np.random.randint(0, 10, 20)
print(a)
print(list(a).count(9))
print(list(a).count(7))
```

```
print(list(a).count(0))
print(np.histogram(a, 10))
```

```
[2 3 8 1 3 3 3 7 0 1 9 9 0 4 7 3 2 7 2 0]
2
3
3
(array([3, 2, 3, 5, 1, 0, 0, 3, 1, 2]), array([0. , 0.9, 1.8, 2.7, 3.6, 4.5, 5.4, 6.3, 7.2, 8.1, 9.
]))
```

In [47]:
```
print(np.max(np.random.randint(0, 10, 5)))
print(np.amax(np.random.randint(0, 10, 5)))
```

```
6
9
```

In [48]:
```
a = np.random.randint(0, 10, 20)
print(a)
bins = np.arange(0, 11)
print(np.histogram(a, bins))
```

```
[8 1 1 7 9 9 3 6 7 2 0 3 5 9 4 4 6 4 4 3]
(array([1, 2, 1, 3, 4, 1, 2, 2, 1, 3]), array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10]))
```

In [49]:
```
a = np.random.randint(0, 10, 20)
print(a)
bins = np.array([0, 2, 4, 6, 8, 10])
print(np.histogram(a, bins))
```

```
[4 4 8 4 3 7 5 5 0 1 5 9 3 0 5 0 1 2 4 2]
(array([5, 4, 8, 1, 2]), array([ 0,  2,  4,  6,  8, 10]))
```

In [57]:
```
a = np.random.randint(0, 9, 20)
print(a)
mask = (a % 2 == 0) & ~(a % 3 == 0)
print(mask)
```

```
[4 5 4 4 8 1 5 7 0 1 3 2 8 2 4 8 2 8 7 8]
[ True False  True  True  True False False False False False False  True  True  True  True  True
True  True False  True]
```

In [59]:
```
print(np.array([[1, 2, 3], [4, 5, 6]]) * 2)
```

```
[[ 2  4  6]
 [ 8 10 12]]
```

In [72]:
```
with open("../var/data.csv") as fh:
        data = csv.reader(fh)
        next(data)
        data = np.array(list(data)).astype(np.float32)
        # velocities = np.array([0.0] * 99)
        # for i in range(1, 100):
        #       velocities[i - 1] = (data[i, 1] - data[i - 1, 1]) / (data[i, 0] - data[i - 1, 0])
        velocities = (data[1:, 1] - data[:-1, 1]) / (data[1:, 0] - data[:-1, 0])
```

```
        print(velocities)
        print(velocities.size)
        minima = np.min(velocities)
        maxima = np.max(velocities)
        average = np.mean(velocities)
        stddev = np.std(velocities)
        print(minima, maxima, average, stddev)
```

```
[50.86623  51.94934  35.591457 67.22993  39.10074  52.93709  48.708946 62.20773  51.321873 31.732359
57.36474  53.031868 60.732754 43.19154  57.944122 34.83306  57.73253  42.088963 57.985405 38.320206
67.48615  48.09369  42.313274 53.712402 47.405327 56.94248  34.93046  69.07307  34.255444 50.61207
47.31065  60.685814 47.955124 52.66938  39.173496 62.67283  47.347755 51.060844 44.845284 57.789318
40.059    59.189426 49.939377 44.040714 53.832962 44.590523 54.019337 57.628765 42.551224 43.362564
66.71175  32.38972  58.073647 51.45259  48.631332 50.34827  44.42236  46.138954 61.216114 43.69419
46.523235 61.169582 45.079113 60.60429  33.92765  59.269466 46.006615 61.938175 46.226128 46.023983
43.85037  55.01503  47.493366 45.73957  47.676678 52.023724 48.457993 52.281235 60.61171  43.871925
42.608994 65.618484 39.051655 64.744255 35.278625 56.09141  55.947453 45.5867   49.73478  57.641296
38.76797  59.92297  29.427637 50.63078  63.78438  51.822433 45.13524  39.2755   49.784782]
99
29.427637 69.07307 49.981247 9.043447
```

In [86]:

```
# print(data.astype(decimal.Decimal))
x = data[:, 0]
y = data[:, 1]
xy = x * y
xx = x * x
n = len(x)
sigmaXY = np.sum(xy)
sigmaX = np.sum(x)
sigmaY = np.sum(y)
sigmaXX = np.sum(xx)
m = ((n * sigmaXY) - (sigmaX * sigmaY)) / ((n * sigmaXX) - (sigmaX ** 2))
print(m)
c = ((sigmaY * sigmaXX) - (sigmaX * sigmaXY)) / ((n * sigmaXX) - (sigmaX **
2))
print(c)
```

```
 1
1
 1
2
 1
3
 1
4
```

```
49.97795
10.081511
```